

A First Project Proposal Report on
**Student Record Management System: A C++ Console-Based Academic
Records Application**

Submitted in Partial Fulfilment of the Requirements for
The Degree of **Bachelor of Engineering in Computer Science**

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ABSTRACT

The “Student Record Management System” is a lightweight, console-based application designed to simplify the process of managing student academic records for small and medium educational institutions. The primary objective of the system is to automate student enrollment, grade computation, and academic reporting, replacing manual registers and unstructured spreadsheets. Educators can enroll students, record marks across multiple subjects, and instantly obtain computed totals, averages, percentages, and letter grades. In many small institutions, the process of maintaining and consolidating student records involves repeated manual calculation and cross-checking, which is time consuming and prone to error. The proposed system intends to eliminate these inefficiencies by automating the process end-to-end.

Additionally, a ranked class report will be generated automatically, streamlining the process of identifying top performers, computing class averages, and tracking pass rates. The system will be implemented in C++ using an object-oriented design, with all student records persisted to a local file so that no data is lost between sessions. With automated grade calculation and consolidated reporting, the process of maintaining academic records is greatly simplified. This project will therefore simplify student record management, solving the problems that exist in the traditional, manual approach.

Keywords: Student, Administrator, Grade Computation, Report Generation, Data Persistence

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1. INTRODUCTION

This project carries the motive of providing an efficient and reliable way to manage student academic records, whether for schools, coaching centers, or small colleges. In this section, the problems, motivations, objectives, project scope, and limitations are described in detail.

1.1 PROBLEM STATEMENT

In the present world, managing student academic records manually can be a challenging and error-prone task, whether for a single teacher or a small administrative office. The conventional approach of maintaining paper registers or unstructured spreadsheets to record marks, compute grades, and generating reports is time-consuming and inefficient.

Also, it becomes difficult for educators to consolidate results across an entire class, rank students, and calculate statistics such as class averages and pass rates. The existing manual approach requires repeated, error-prone arithmetic and cross-checking, which increases the likelihood of mistakes in the final reports issued to students and guardians.

This can lead to frustration, wastage of time and effort, and difficulties in producing accurate, timely academic reports.

1.2 PROJECT OBJECTIVES

To address the problems stated above in the problem statement (section 1.1), the existing manual approach to student record keeping should be reformed. So, with the purpose of providing an efficient solution to the problems stated, the Student Record Management System is proposed with the following objectives:

- i. Simplify student record keeping: The primary objective of this project is to provide an efficient, structured platform for enrolling students and recording their academic marks.
- ii. Automate grade computation: Functioning as an automated system, it computes totals, averages, percentages, and letter grades without manual arithmetic.
- iii. Increase efficiency and save time: Unlike conventional methods, educators do not have to manually calculate or cross-check results. The system generates individual and class-wide reports instantly.
- iv. Eliminate manual error: The traditional approach is highly susceptible to arithmetic and transcription errors. This system focuses on eliminating them through automated, validated computation.

1.3 SCOPE AND LIMITATIONS

In the project ‘Student Record Management System’, a console-based application will be developed in such a way that an administrator or educator can enroll, update, search, and report on student records through a centralized, file-persistent platform.

The scopes of this project are:

- v. The target audience can be any educator, small institution, or coaching center that needs to manage student academic records.
- vi. It especially targets small to medium class sizes where a full database-backed system would be unnecessary overhead.
- vii. The system can be used to track enrollment, marks, and generate reports for any subject configuration within the fixed five-subject model.

The limitations of this project are:

- viii. It is currently limited to a fixed set of five subjects per student.
- ix. The system supports a single user at a time and does not provide authentication or role-based access as of now.

2. LITERATURE REVIEW

In this section, existing tools and approaches that provide similar student record management functionality are reviewed.

Many institutions currently rely on generic spreadsheet software such as Microsoft Excel or Google Sheets to record student marks and compute grades. While flexible, these tools require manually written formulas for grade computation and ranking, which are easy to break through accidental edits and offer no built-in validation of entered marks. Furthermore, spreadsheets do not enforce a consistent record structure, which can lead to inconsistent or incomplete student data over time. In our proposed system, structured data entry and automatic validation eliminate the risk of malformed records and broken formulas.

Dedicated school-management platforms such as Fedena and OpenSIS also provide student record and grading functionality. However, these are typically web-based, multi-user systems that require server installation, database configuration, and ongoing maintenance overhead that is unnecessary for a single educator or a very small institution. In our proposed system, a lightweight, dependency-free, console-based application is used instead, backed up by simple local file storage, making it far easier to deploy and use without any server or network setup.

3. METHODOLOGY

The following methodology will be adopted for a smooth development process, ensuring that the project meets requirements and does not deviate from its scope.

3.1.1 REQUIREMENT ANALYSIS

This is the first step of the incremental model, where requirements are collected and analyzed. The outcome of this phase is a System Requirement Specification (SRS) document.

3.1.2 DESIGN

In this phase, the system is designed based on the System Requirement Specification (SRS). A class diagram and a case diagram will be developed in this phase.

3.1.3 CODING

The functionalities specified in the SRS and modeled during the design phase are implemented in this phase. The outcome is a working product with the functionalities specified in the document.

3.1.4 TESTING AND EVALUATION

In this phase, individual modules of the system are tested separately. The entire system is also tested end-to-end to ensure that all functionalities work smoothly as specified.

3.2 TOOLS TO BE USED

The following tools will be used throughout this project to streamline the development process.

TRELLO:

Trello is a web-based project management tool that will be used to organize and track individual development tasks.

GIT AND GITHUB:

Git is a distributed version control system used to manage different versions of the source code. GitHub will be used to host the project repository and track changes over time.

VS CODE:

VS Code will be used as the primary code editor for development tasks.

3.3 TECHNOLOGIES TO BE USED

The following technologies will be used during project development.

C++:

C++ is a general-purpose, compiled programming language that supports object-oriented design. It will be used to implement the entire application, including the data model and business logic.

STANDARD TEMPLATE LIBRARY (STL):

STL provides data structures such as vector and algorithms such as sort, which will be used to manage the in-memory collection of student records and generate ranked reports.

FILE I/O (fstream):

The C++ Standard Library's file stream classes will be used to persist student records to, and load them from, a local flat file.

4. SYSTEM DESIGN

This section provides a comprehensive overview of the proposed system's architecture, focusing on two key diagrams: the class diagram and the use case diagram. These diagrams illustrate the core components and the interactions between them in the system.

4.1 CLASS DIAGRAM

A class diagram is a visual representation of the classes within a system, their attributes and methods, and the relationships between them. The major classes involved in our system are Student and Student Management System. The relationship between these classes, along with the persisted data file, is shown in the diagram below.

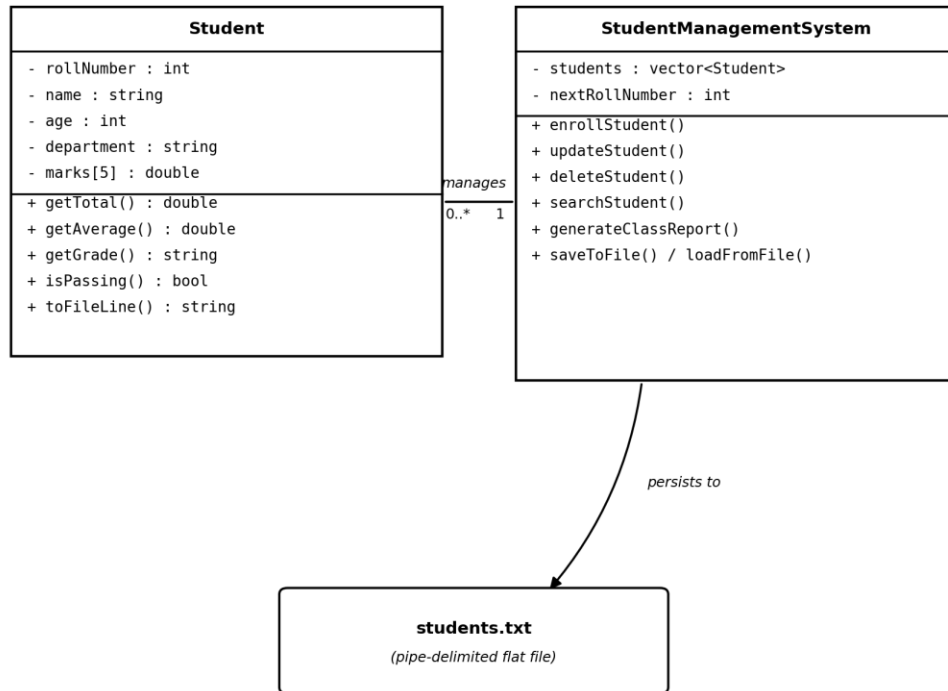


Figure 2: Class Diagram

4.2 USE CASE DIAGRAM

A use case diagram is a visual representation of the interactions between an actor (in our case, the User/Administrator) and the system under consideration. It shows the different use cases or functionalities provided by the system and the relationships between the actor and the use cases.

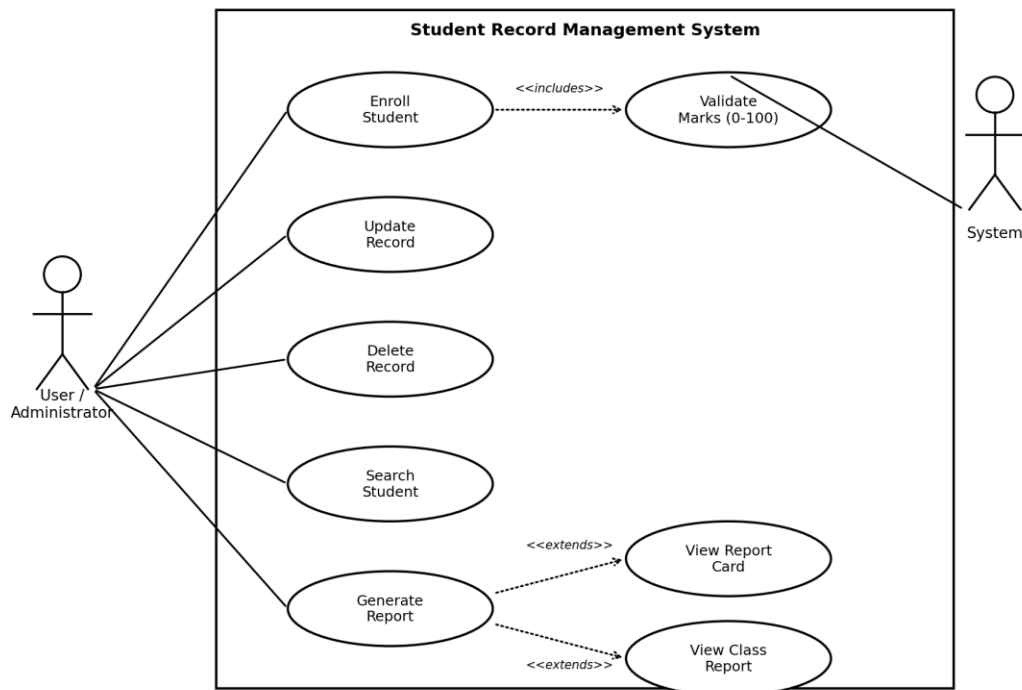


Figure 3: Use Case Diagram

4.3 SEQUENCE DIAGRAM

A sequence diagram illustrates the order in which messages are exchanged between the actor and the system's objects to complete a task over time. The diagram below shows the enrollment workflow, from the User/Administrator invoking enrollment through grade computation and persistence to the students.txt file, and finally report generation.

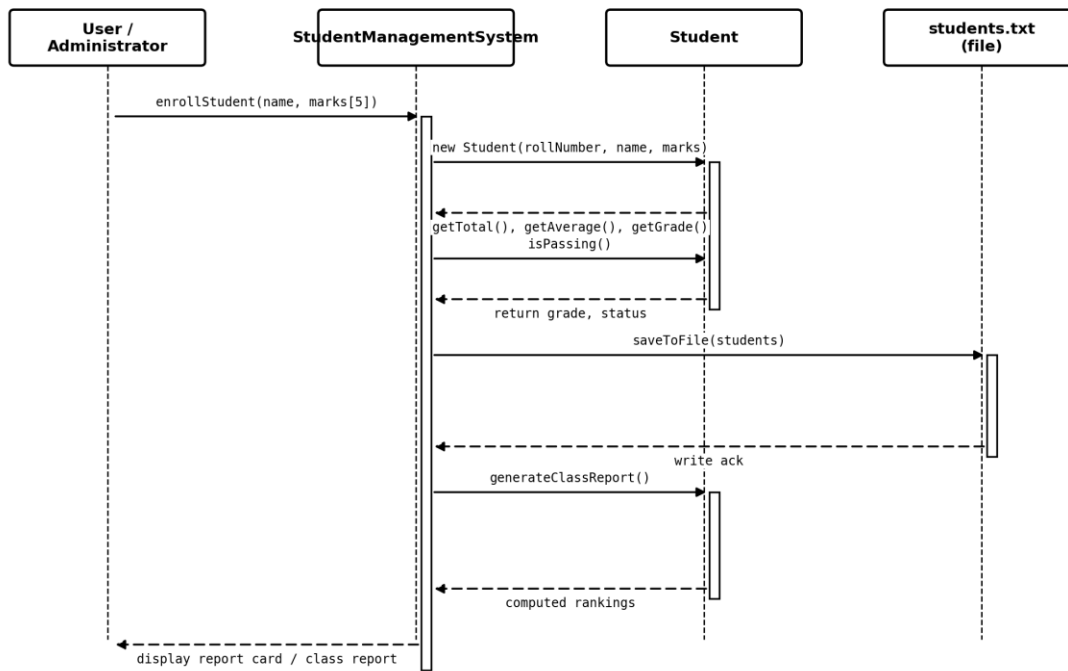


Figure 4: Sequence Diagram

5. PROPOSED DELIVERABLES

The project will deliver a lightweight, console-based Student Record Management System written in C++. The user will be able to run and use the application directly from the command line without the need for any external server or database setup. There is essentially one primary user of the application: the Administrator or Educator, who manages student enrollment, records, and reports.

In the application, the administrator is able to enroll students, enter and validate marks, and automatically compute grades. The administrator can also search, update, and delete existing student records.

The system will generate individual report cards showing subject-wise marks, totals, averages, percentages, and grades, as well as a ranked, class-wide report showing class averages, highest and lowest scores, and the overall pass rate.

6. PROJECT TASK AND TIME SCHEDULE

The project time schedule is designed considering the requirements and constraints involved. The schedule is designed such that the Requirement Analysis and Core Development phases are given more importance so that the project does not deviate from its core requirements.

Table 1: Time Schedule

Task	Approx. Duration (Days)
Requirements Analysis	5
System Design	3
Core Development (Classes)	6
Persistence Layer	3
Reporting Module	4
Testing and debugging	4
Documentation	5

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